Particle System Usage Manual for UI

This manual provides guidance on correctly using particle systems in UI, avoiding common issues, and optimizing particle effects.

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1. Basic Considerations

1.1 Canvas RenderMode Settings

When using particle systems in UI, the **Canvas RenderMode must not be set to Overlay**, as this mode prevents proper rendering of particle systems. The following modes are recommended:

- **Screen Space - Camera**: Binds the particle system to the camera, supporting depth sorting.
- **World Space**: Keeps the particle system in world coordinates along with other UI elements, making layer adjustments easier.

1.2 Display Layer Relationship

The display order between the UI and particle systems can be adjusted using the following methods:

1. **Modify Z-axis coordinates**: Adjust the Z-axis position of the particle system or UI elements to control their front-to-back relationship.
2. **Use SortingLayer**: Create a separate Canvas for UI elements or the particle system, and configure the related SortingLayer properties to ensure the particle system appears at the correct layer.

Tip: Refer to example scenes included in the project for detailed setup guidance.

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2. Using Layers and RectTransform

- **Layer Settings**: Ensure the particle system's Layer matches the Canvas Layer to avoid rendering issues.
- **RectTransform Settings**: When attaching particle systems to UI, adjust position and size through the `RectTransform` component to align with the Canvas.

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3. Key Parameter Descriptions

3.1 Scaling Mode

The `Scaling Mode` determines how the particle system behaves under different scaling conditions:

- **Hierarchy**: The particle size scales with its parent object.
- **Local**: The particle size remains fixed and unaffected by parent object scaling.
- **Shape**: Only the emission area scales; the particle size itself remains unchanged.

For UI, adjust the `Scaling Mode` based on Canvas requirements to ensure particle sizes meet expectations.

3.2 Renderer Component Parameters

- **Min Particle Size**: Specifies the minimum display size of particles. Increasing this value prevents particles from becoming too small to be visible.
- **Max Particle Size**: Specifies the maximum display size of particles, preventing particles from obstructing other UI elements.

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4. `ParticleSystemMoveOnEdge` Script

This script enables particle systems to move along UI edges. Key considerations include:

- Most script parameters are automatically retrieved from the particle system, such as emission direction and color. Modify the particle system directly for changes.
- Detailed comments are included in the script; review them for guidance on parameter adjustments.

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5. Multiple Effect References

The example scenes provide various particle effects, showcasing different parameter applications. By comparing these effects, you can learn how to adjust the following features:

- **Particle Color Gradients** (`Color Over Lifetime`)
- **Emission Shapes** (`Shape`)
- **Particle Speed and Direction** (`Start Speed / Velocity Over Lifetime`)

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6. Common Parameters Affecting Particle Effects

6.1 Emission Module

- **Rate Over Time**: Number of particles emitted per second; ideal for stable effects.
- **Bursts**: Emits a group of particles at specific intervals; suitable for explosion-like effects.

6.2 Shape Module

- **Shape Type**: Defines the emission area, such as circle, cone, or edge.
- **Radius/Angle**: Adjusts the range and direction of the emission area.

6.3 Size Over Lifetime

Controls particle size changes over their lifetime, useful for simulating shrink or growth effects.

6.4 Color Over Lifetime

Gradually changes particle colors over time, ideal for dynamic light effects or smoke.

6.5 Velocity Over Lifetime

Modifies particle speed over time, great for creating flowing effects.

6.6 Noise Module

Adds random motion to particles, simulating natural effects like flickering flames or rippling water.

7. Debugging and Optimization Suggestions

- **Max Particles**: Limit the number of particles in UI to improve performance.
- **Prewarm**: Enables the particle system to display a fully developed effect upon start.
- **Culling Mode**: Enable culling based on actual needs to save resources.

By carefully configuring and optimizing parameters, you can design particle effects that perform exceptionally well in UI environments!